Group Theory

Let G be a set.

- Binary operation $\bullet : G^2 \to G$.

  $\forall\, a, b \in G,\quad a \cdot b = c$ and $c \in G$.

  "Closure": Operation yields an element that already $\in G$.

Examples: 1. $G = \mathbb{Z}$ and $\bullet = +$

$\forall\, a, b \in \mathbb{Z},\quad a+b \in \mathbb{Z}$.

2. $G = \mathbb{Z}_n,\quad \bullet = +$ mod n

- Group: $(\overset{\text{set}}{G}, \underset{\text{binary operation}}{\bullet})$

Let G be a set together with a binary operation that assigns an ordered pair of elements in G (say (a,b)), an element $a \cdot b$ in G. We say G is a "group" under this operation if the following properties hold.

1. Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)\ \forall\, a, b, c \in G$.
2. Identity: There is an element $e \in G$ s.t $\forall\, a \in G$, $a \cdot e = e \cdot a = a$.
3. Inverses: For all elements a in G, there is an

element $b \in G$ s.t $ab = b \cdot a = e$.

<u>Remark</u>: 1. We shall soon drop the notation $a \cdot b$ and start using ab instead.

2. It is not always necessary that $ab = ba$.
↳ If that is true for all $a, b \in G$, then G is "Abelian".

Examples:

1. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are all groups under addition
↳ Check for associativity, identity and inverse.

2. $\mathbb{Z}$ is not a group under multiplication
↳ for $2 \in \mathbb{Z}$, its inverse under mult. $\frac{1}{2}$ does not belong to $\mathbb{Z}$.

3. Subset $\{1, -1, i, -i\}$ is a group under complex mult.

4. $\mathbb{Q}^+$ is a group under ordinary mult.

5. $S = \{$positive irrational numbers$\} \cup \{1\}$ is not a group as $\sqrt{2} \cdot \sqrt{2} = 2$ and $2 \notin S$.

6. Set of all 2×2 matrices with real entries is a group under component wise addition.
↳ $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is the identity.

$\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ for $n \geq 1$ is a group under addition mod n.

$\forall j$, $n-j$ is the additive inverse modulo n.

8. $\mathbb{R}^*$: Set of all non-zero real no.s is a group under ordinary mult.

Identity $e = 1$. $\forall a$, $\frac{1}{a} \in \mathbb{R}^*$.

9.

$$GL(2, \mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \middle| \begin{array}{l} a, b, c, d \in \mathbb{R} \text{ and} \\ ad - bc \neq 0 \end{array} \right\}.$$

All 2x2 matrices with real entries and non-zero determinant form a group under matrix multiplication

Identity: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Inverse: $\begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$

Useful fact:

$\det(A^{-1}) = (\det(A))^{-1}$.

Remark: 1. $GL(2, \mathbb{R})$ is an example of a non-Abelian group.

definition can be extended to arbitrary $n \times n$ matrices.

2. If $ad = bc$ then matrix inverse may not exist and thus all 2x2 matrices with Real entries do not form a group under matrix multiplication.

10. $U(n) := \{ a \mid a \leq n \text{ and } \gcd(a,n) = 1 \}$

Set of all positive integers less than n that are relatively prime ~~with~~ to n.

$U(n)$ is a group under multiplication modulo n.

<u>Ex</u>: for $n = 10$, $U(n) = \{1, 3, 7, 9\}$.

mod 10	1	3	7	9
1	1	3	7	9
3	3	9	1	7
7	7	1	9	3
9	9	7	3	1

7×9
$= 63$.
$63 \equiv 3 \bmod 10$

A table representation of binary operation on group elements is called Cayley table.

<u>Remark</u>: If n is prime, $U(n) = \{1, 2, \ldots, n-1\}$.

11. Set of integers under subtraction is not a group

Associativity fails.

<u>Ex</u>: $(5-3)-2 \neq 5-(3-2)$

ymmetries of a square:

$R_0 :=$

P	W
G	B

$\longrightarrow$

P	W
G	B

$R_{90} :=$

P	W
G	B

$\longrightarrow$

W	B
P	G

$R_{180} :=$

P	W
G	B

$\longrightarrow$

B	G
W	P

$R_{270} :=$

P	W
G	B

$\longrightarrow$

G	P
B	W

$H :=$

P	W
G	B

$\longrightarrow$

G	B
P	W

flip across horizontal axis

$V :=$

P	W
G	B

$\longrightarrow$

W	P
B	G

flip across vertical axis

$D =$

P	W
G	B

$\longrightarrow$

P	G
W	B

flip across main diagonal.

$D' =$

P	W
G	B

$\longrightarrow$

B	W
G	P

flip across the other diagonal.

Ex: $R_{90} \xrightarrow{\text{then}} H \equiv D$.

Mathematical way...
$H(R_{90}) \equiv D$.

$D_4 = \{R_0, R_{90}, R_{180}, R_{270}, H, V, D, D'\}$

form a group under composition of operations.

$\forall\, a, b \in D_4$, $ab \in D_4$ too.

R_0 is identity.

HW: Figure out inverses of these operations

Qn: Is this Abelian?

↳ No! $HD \neq DH$.

Preliminary lemmas:

<u>Lemma</u>: If G is a group, then

1. The identity element e of G is unique.
2. Every $a \in G$ has a unique inverse in G.
3. For every $a \in G$, $(a^{-1})^{-1} = a$.
4. For all $a, b \in G$, $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$.

0. If $ba = ca$ then $b = c$. Similarly, $ab = ac \Rightarrow b = c$.

<u>Pf</u>:

0: If $ba = ca$ then $(ba)a^{-1} = (ca)a^{-1}$

$$\Rightarrow b(aa^{-1}) = c(aa^{-1}) \quad \text{associativity}$$

$$\Rightarrow be = ce$$

$$b = c.$$

Similarly for $ab = ac$.

$$a^{-1}(ab) = a^{-1}(ac)$$

$$(a^{-1}a)b = (a^{-1}a)c \quad \text{associativity.}$$

$$eb = ec \Rightarrow b = c.$$

1. Say both e and f are both identities s.t $ea = a$ and $fa = a$. Further assume $e \neq f$.

$e \cdot a = a$ and $e \cdot f = f$ similarly $e \cdot f = e$

Then $f = e \cdot f = e$.

2. Let x and y be such that

$$a \cdot x = a \cdot y = e.$$

$ax = ay$ suppose there is an elem $b \in G$ s.t

$b \cdot a = e$ then $b \cdot (ax) = b \cdot (a \cdot y)$.

$$x = e \cdot x = (b \cdot a) \cdot x = b(a \cdot x) = b \cdot (a \cdot y) = (b \cdot a) \cdot y$$
$$= e \cdot y = y.$$

3. $a^{-1} \cdot (a^{-1})^{-1} = e = a^{-1} \cdot a$

$$\Downarrow$$

$$a^{-1} \cdot (a^{-1})^{-1} = a^{-1} \cdot a \Rightarrow (a^{-1})^{-1} = a.$$

4. $(a \cdot b) \cdot (b^{-1} a^{-1}) = a \cdot (b \cdot b^{-1}) a^{-1} = a \cdot (e \cdot a^{-1})$

$$= a \cdot a^{-1} = e.$$

$$\Rightarrow (a \cdot b)^{-1} = b^{-1} \cdot a^{-1}.$$

Subgroup:

<u>Defn</u>: A non-empty subset H of G is said to be a subgroup of G if under the binary operation in G, H itself forms a group.

ormally, a non-empty subset H of G is a sub group of G ~~then~~ if and only if

1. $a, b \in H \Rightarrow ab \in H$

2. $a \in H \Rightarrow a^{-1} \in H$.

<u>Lemma</u>: If H is a nonempty finite subset of G and H is closed under binary operation, then H is a subgroup of G.

<u>Pf</u>: Since H is closed under the group operation, for all a, $a \cdot a = a^2 \in H$ as well. Further for all $i \in \mathbb{N}$, $a^i \in H$. But our assumption was that H is finite. $\Rightarrow$ There must be repetition of elems. That is for some $r > s > 0$, $a^r = a^s$.

By cancellation, $a^{r-s} = e$.

$\downarrow$

This ensures that $e \in H$.

Further, $(a \cdot a^{r-s-1}) = e \Rightarrow (a^{r-s-1})$ is a^{-1}. $\square$

$\Rightarrow$

Sufficient criterion: To check if H is a subgroup of G, it is sufficient to check if H is closed under the group operation.

Ex: 1. multiples of 5 form a subgroup of $\mathbb{Z}$,

2. $H = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \;\middle|\; \begin{matrix} ad \neq 0 \\ a,b,d \in \mathbb{R} \end{matrix} \right\}$ is a subgroup of $GL(n, \mathbb{R})$ under mat mult.

3. $H = \{a + bi \mid a^2 + b^2 = 1\}$ is a subgroup of $\mathbb{C}$ under $+$.

Defn: Let G be a group, H be a subgroup of G. For $a, b \in G$, we say $a \equiv b \mod H$ if $ab^{-1} \in H$.

Lemma: $a \equiv b \mod H$ is an equivalence relation.

Pf: We need to show $\forall\, a, b, c \in G$,

1. $a \equiv a \mod H$
2. $a \equiv b \mod H \Rightarrow b \equiv a \mod H$
3. $a \equiv b \mod H,\ b \equiv c \mod H \Rightarrow a \equiv c \mod H$.

1. Since H is a subgroup, $e \in H \Rightarrow a \cdot a^{-1} \in H$.

$\Rightarrow a \equiv a \mod H$.

2. $a \equiv b \mod H \Rightarrow ab^{-1} \in H$. Further, $(ab^{-1})^{-1} \in H$.

$\Rightarrow (ab^{-1})^{-1} = (b^{-1})^{-1} a^{-1} = ba^{-1} \in H$.

Thus $b \equiv a \mod H$.

$a \equiv b \bmod H$ and $b \equiv c \bmod H$

$\Rightarrow$ $ab^{-1} \in H$ and $bc^{-1} \in H$.

Product of two elems in H must be in H.

$$ab^{-1}bc^{-1} = a(b^{-1}b)c^{-1} = ac^{-1} \in H$$

$$\Rightarrow a \equiv c \bmod H.$$

<u>Remark</u>: If $G = \mathbb{Z}$ and $H = H_n$ multiples of n then

$\cdot$ = addition

$a \equiv b \bmod H \Rightarrow a - b \in H$

additive inverse.

That is, a and b are equivalent modulo n.

<u>Defn</u>: If H is a subgroup of G, $a \in G$ then $Ha = \{ha \mid h \in H\}$. Ha is called a right coset of H in G.

<u>Lemma</u>: For all $a \in G$,

$$Ha = \{x \in G \mid a \equiv x \bmod H\}.$$

<u>Pf</u>: Let us use $[a]$ to represent the set

$$\{x \in G \mid a \equiv x \bmod H\}.$$

First, we will show that $Ha \subseteq [a]$. That is, every element of Ha is in $[a]$,

Note that if $h \in H$ then $h^{-1} \in H$.

$$(a \cdot a^{-1}) h^{-1} \in H \Rightarrow (a(a^{-1} \cdot h^{-1})) \in H$$

$$\Downarrow$$

$$a \cdot (ha)^{-1} \in H$$

$$\Rightarrow$$

$$a \equiv ha \mod H.$$

$$\Rightarrow ha \in [a].$$

OTOH, $x \in [a] \Rightarrow ax^{-1} \in H \Rightarrow (ax^{-1})^{-1} \in H$

$$\downarrow$$

$$xa^{-1} \in H.$$

$\downarrow$ multiplying by a on both sides

$$xa^{-1}a \in Ha.$$

$$\Rightarrow x \in Ha.$$

$$\Rightarrow Ha = [a].$$

<u>Remark</u>: The above lemma states that Ha forms an equivalence class.

Further, we will study that two cosets Ha and Hb are either identical or have no element in common.

$\rightarrow$

<u>Lemma</u>: There is a one-to-one correspondence between any two right cosets of G.